# HOTEL BOOKING ANALYSIS

INSIGHTS REPORT

Excel • Power Query • Pivot Tables • Dashboard

## Project overview

This project explores hotel booking data for City Hotel and Resort Hotel to understand booking volume, cancellations, room preferences, guest types, and monthly patterns. It was built as a practical Excel analytics project using Power Query, formulas, Pivot Tables, and an interactive dashboard.

| 37.04% | 41.73% | 27.76% |
|---|---|---|
| Overall cancellation rate | City Hotel cancellation rate | Resort Hotel cancellation rate |

## Tools and workflow

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|---|---|
| **Power Query** | Loaded, cleaned, and transformed the dataset; removed unnecessary columns. |
| **Excel formulas** | Used IF/AND logic to classify room preference as Desired/Un-Desired and guest type as Couples/Family/Singles. |
| **Pivot Tables** | Summarized booking and cancellation patterns across categories and time. |
| **Dashboard** | Presented KPIs and charts with a year slicer for interactive exploration. |

## Key findings

**Cancellations are substantial.** The overall cancellation rate is 37.04%, meaning a little over one-third of bookings were cancelled.

**City Hotel has a higher cancellation rate.** City Hotel is at 41.73%, compared with 27.76% for Resort Hotel—a difference of 13.97 percentage points.

**August has the highest cancelled-booking count.** August recorded 5,239 cancelled bookings in the analysis.

**Couples show the highest cancellation rate among guest types.** The cancellation rate for Couples is 39.75%.

**Room preference is associated with cancellation outcomes.** Bookings classified as Desired Room have a 41.56% cancellation rate, while Un-Desired Room bookings have a 5.38% rate.

# Business recommendations

### 1. Review City Hotel cancellation patterns

Investigate booking lead time, deposit/payment type, customer segment, and booking channel for City Hotel. Use the results to test targeted confirmation reminders or suitable deposit policies.

### 2. Prepare for seasonal cancellation peaks

Because August has the highest cancelled-booking count, review the month's booking mix and cancellation timing. Plan staffing and inventory controls around observed seasonal patterns.

### 3. Improve booking confirmation for couples

Test clear confirmation messages and pre-arrival reminders for bookings classified as Couples. Compare cancellation rates before and after the intervention.

### 4. Validate room-preference classification

The gap between Desired Room and Un-Desired Room cancellation rates is worth investigating. Confirm how these labels were assigned and check whether other booking characteristics explain the difference.

### 5. Track cancellation rate alongside volume

Monitor both the number of cancellations and cancellation rate by hotel, month, and guest type. Counts alone can be misleading when booking volumes differ.

# Conclusion

The analysis highlights differences in cancellation behavior across hotel types, months, guest types, and room-preference categories. The dashboard makes these patterns easier to explore and suggests where hotel teams could investigate further. A business team should validate the patterns with operational context and test targeted actions against measurable outcomes.

# Skills demonstrated

| | | | |
|---|---|---|---|
| Data cleaning | Excel formulas | Pivot analysis | Dashboard design |